

Technical Document #1008: Retrieval Augmented Generation - Comprehensive Overview

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.
- Re-ranking models score retrieved documents for relevance to the query.
- Vector databases store dense embeddings for semantic search.